

Dhruv Desai

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EDUCATION

University of Wisconsin - Madison at Madison, WI GPA - 4.0/4.0
Bachelor of Science in Computer Science (Honors) & Math for Programming and Computing Sep 2022 - May 2026
Business, Certificate Sep 2024 - May 2026
Coursework *Linear Algebra, Probability, Stochastic Processes, Linear Optimization, Math Methods in Data Science Algorithms (Honors), Operating Systems, Machine Organization, Computer Engineering, Compilers, AI*
Self Study: *Stanford's Language Modeling from Scratch, MIT's Distributed Systems*

EXPERIENCE

Software Dev Eng (DB) at Amazon Web Services (ElastiCache), Seattle, WA Sep 2026 – Present

- Working on ElastiCache's core data-plane engine, owning crash analysis and production deep dives
- Mitigated a TLS connection storm by experimenting with and retuning the engine's connection-queue sizing
- Navigated ~400M log lines with agentic AI tooling to identify and root-cause a security vulnerability

Software Engineer at Commvault, Hybrid May 2025 – Sep 2026
Software Engineer (Full Time)

- Rewrote a legacy service in Rust via an autonomous agentic pipeline, validated by shadowing production traffic
- Cut per-request latency ~30s to 700ms and build times ~8h to 1.5h, saving ~\$1,700/yr in licensing
- Co-owned UDV, Commvault's S3-compatible storage interface, end to end, resolving operational pains and bugs
- Built backup & restore for the Monday.com platform on Commvault infrastructure using an agentic AI coding assistant

Software Engineering Intern

- Built a cipher-detection ML model with 80% accuracy & 0.79 F1-score using XGBoost, CNNs, and LightGBM
- Integrated C++ extractors with an ONNX inference pipeline, achieving 270ms latency for 40KB inputs
- Managed build tool reliability, resolving 20+ critical incidents via server log analysis and root cause remediation

Research Assistant (Honors Thesis) at University of Wisconsin-Madison, Madison, WI Jan 2025 – May 2026

- Authored thesis *MemEvolve* (advisor: Prof. Michael Swift): used LLMs to evolve Linux page-cache eviction policies as eBPF
- Deployed evolved policies live in a custom cache_ext 6.6.8 kernel via eBPF struct_ops, with no rebuild or simulator
- Built a breadth-first evolutionary search (planner/mutator/frontier-reviewer) refining policies from kernel-probe feedback
- Engineered a distributed evaluator across CloudLab nodes, isolating each candidate in a cgroup to remove cache interference
- Achieved 18% higher throughput on scan-based workloads; workload-specialized policies beat cache_ext's baselines

Course Assistant – OS, AI at University of Wisconsin-Madison, Madison, WI Sep 2023 – May 2025

- Taught systems programming in C: pointers, manual memory management, and concurrency for OS kernel projects
- Mentored 600+ students, debugging their C and kernel-space code and resolving 98% of implementation issues
- Authored a GDB-based guide for debugging concurrent C programs in kernel space, adopted by TAs for 2 semesters
- Led 10–15 student system design sessions on COW, huge pages, RAID, lock-free buffers, and cache coherence

PROJECTS

MyTorch | C++, CUDA, GPU Kernels, Autograd, FlashAttention, Diffusion | [Gallery](#) Present

- Built the entire deep-learning stack in C++/CUDA with zero ML dependencies: reference-counted storage, strided N-d Tensors with zero-copy views and broadcasting, a CPU/CUDA dispatch layer, reverse-mode autograd, and nn modules
- Drove SGEMM from 7.5 to ~63 TFLOP/s (~80% of cuBLAS) on RTX 5090 in 7 Nsight-guided stages of shared-memory tiling, 8x8 register tiling, float4 vectorization, warp tiling, and double buffering, fixing the top bottleneck each time
- Implemented a FlashAttention kernel with online softmax and SRAM tiling, avoiding the $O(n^2)$ attention matrix in HBM
- Trains real generative models end to end: a Llama-3.1-style transformer and a UNet diffusion image generator
- Verified the analytical autograd engine by checking every gradient against numerical finite-difference estimates
- Used AI as a Socratic teacher: the repo's CLAUDE.md has agents explain, review, and question, never write the code

PrimitiveDB | C++, Systems Design, PAX Storage, Cache Locality, perf_event_open March 2026

- Built a C++ harness benchmarking PAX columnar vs row-major storage and radix vs naive joins via perf_event_open(2)
- Sped single-column scans 6x and cut L1D misses ~14.5x via a PAX columnar layout packing 16 values per cache line
- Cut join latency 2.7x and L3 misses ~170x with a cache-conscious radix-partitioned hash join sized to L1/L2

TECHNICAL SKILLS

Languages: C, C++, CUDA C/C++, Python, Rust, Go
GPU & Parallel Computing: CUDA, Triton, GPU kernels, Tensor Cores, cuBLAS, FlashAttention, PTX/SASS, tiling
Performance & Profiling: Nsight Compute (roofline), perf, Valgrind, occupancy, coalescing, bank-conflict analysis
Math & Numerical Methods: Linear Algebra, Numerical Methods, Numerical Stability, Mixed Precision, Probability
ML & DL: PyTorch, Transformers, Diffusion (UNet), CNNs, RNNs/LSTMs, backprop, Adam/SGD, RL, ONNX, XGBoost
Systems & Tools: Linux, eBPF, GDB, CMake, GoogleTest, Git, Docker